

LEVEL 6 — Mathematical Operations

Important for Data Analytics — easy line-by-line examples

First: import NumPy

```
import numpy as np
```

Part A — Basic Arithmetic

1. Addition (+)

Adds corresponding values.

```
a = np.array([10,20,30])  
b = np.array([1,2,3])  
print(a + b)  
Output: [11 22 33]
```

2. Subtraction (-)

Subtracts corresponding values.

```
print(a - b)  
Output: [ 9 18 27]
```

3. Multiplication (*)

Multiplies corresponding values.

```
print(a * b)  
Output: [10 40 90]
```

4. Division (/)

Divides corresponding values. Result is normally floating-point.

```
print(a / b)  
Output: [10. 10. 10.]
```

5. Modulus (%)

Returns the remainder.

```
a=np.array([10,21,32]); b=np.array([3,4,5])  
print(a % b)  
Output: [1 1 2]
```

6. Power (**)

Raises values to a power.

```
a=np.array([2,3,4])  
print(a ** 2)  
Output: [ 4 9 16]
```

7. Floor division (//)

Divides and rounds toward negative infinity.

```
a=np.array([10,11,12]); b=np.array([3,3,5])  
print(a // b)  
Output: [3 3 2]
```

Part B — NumPy Functions

8. np.add()

NumPy function for element-wise addition.

```
print(np.add(a, b))
```

9. np.subtract()

NumPy function for element-wise subtraction.

```
print(np.subtract(a, b))
```

10. np.multiply()

NumPy function for element-wise multiplication.

```
print(np.multiply(a, b))
```

11. np.divide()

NumPy function for element-wise division.

```
print(np.divide(a, b))
```

12. np.mod()

NumPy function for element-wise remainder.

```
print(np.mod(a, b))
```

13. np.power()

Raises each element to a power.

```
a=np.array([2, 3, 4])
```

```
print(np.power(a, 2))
```

Output: [4 9 16]

14. np.sqrt()

Finds the square root.

```
a=np.array([4, 9, 16, 25])
```

```
print(np.sqrt(a))
```

Output: [2. 3. 4. 5.]

15. np.square()

Finds the square of each value.

```
a=np.array([2, 3, 4, 5])
```

```
print(np.square(a))
```

Output: [4 9 16 25]

16. np.abs()

Returns absolute values, removing negative signs.

```
a=np.array([-10, -5, 0, 5, 10])
```

```
print(np.abs(a))
```

Output: [10 5 0 5 10]

17. np.sign()

Returns -1 for negative, 0 for zero, and 1 for positive.

```
a=np.array([-10,0,25])
```

```
print(np.sign(a))
```

Output: [-1 0 1]

Quick Revision

+ Addition | - Subtraction | * Multiplication | / Division | % Modulus | ** Power | // Floor division

np.sqrt() Square root | np.square() Square | np.abs() Absolute value | np.sign() Sign